

# SAFETY DATA SHEET

Revision Date 29-March-2024

Revision Number 5

## 1. Identification

**Product Name** Aluminum perchlorate nonahydrate

**Cat No. :** 11659

**CAS-No** 81029-06-3  
**Synonyms** No information available

**Recommended Use** Laboratory chemicals.  
**Uses advised against** Food, drug, pesticide or biocidal product use.

### Details of the supplier of the safety data sheet

#### Company

##### **Importer/Distributor**

Fisher Scientific  
112 Colonnade Road,  
Ottawa, ON K2E 7L6,  
Canada  
Tel: 1-800-234-7437

#### **Emergency Telephone Number**

For information **US** call: 001-800-227-6701 / **Europe** call: +32 14 57 52 11

Emergency Number **US**:001-201-796-7100 / **Europe**: +32 14 57 52 99

**CHEMTREC** Tel. No. **US**:001-800-424-9300 / **Europe**:001-703-527-3887

## 2. Hazard(s) identification

### Classification

**WHMIS 2015 Classification** Classified as hazardous under the Hazardous Products Regulations (SOR/2015-17)

|                                                         |              |
|---------------------------------------------------------|--------------|
| <b>Oxidizing solids</b>                                 | Category 2   |
| <b>Skin Corrosion/Irritation</b>                        | Category 1 B |
| <b>Serious Eye Damage/Eye Irritation</b>                | Category 1   |
| <b>Specific target organ toxicity (single exposure)</b> | Category 3   |
| Target Organs - Respiratory system.                     |              |

### Label Elements

#### **Signal Word**

Danger

#### **Hazard Statements**

May intensify fire; oxidizer

Causes severe skin burns and eye damage

May cause respiratory irritation



### Precautionary Statements

#### Prevention

Keep away from heat, hot surfaces, sparks, open flames and other ignition sources. No smoking

Keep/Store away from clothing/combustible materials

Take any precaution to avoid mixing with combustibles

Do not breathe dust/fumes/gas/mist/vapours/spray

Wash face, hands and any exposed skin thoroughly after handling

Use only outdoors or in a well-ventilated area

Wear protective gloves/protective clothing/eye protection/face protection

#### Response

IF SWALLOWED: Rinse mouth. Do NOT induce vomiting

IF ON SKIN (or hair): Take off immediately all contaminated clothing. Rinse skin with water/ shower

IF INHALED: Remove person to fresh air and keep comfortable for breathing

IF IN EYES: Rinse cautiously with water for several minutes. Remove contact lenses, if present and easy to do. Continue rinsing

Immediately call a POISON CENTER/doctor

Wash contaminated clothing before reuse

In case of fire: Use dry sand, dry chemical or alcohol-resistant foam to extinguish

#### Storage

Store locked up

Store in a well-ventilated place. Keep container tightly closed

#### Disposal

Dispose of contents/container to an approved waste disposal plant

## 3. Composition/Information on Ingredients

| Component                        | CAS-No     | Weight % |
|----------------------------------|------------|----------|
| Aluminum perchlorate nonahydrate | 81029-06-3 | <=100    |
| Perchloric acid, aluminum salt   | 14452-39-2 | 0        |

## 4. First-aid measures

#### General Advice

Show this safety data sheet to the doctor in attendance. Immediate medical attention is required.

#### Eye Contact

Rinse immediately with plenty of water, also under the eyelids, for at least 15 minutes. Immediate medical attention is required. Keep eye wide open while rinsing.

#### Skin Contact

Wash off immediately with soap and plenty of water while removing all contaminated clothes and shoes. Call a physician immediately.

#### Inhalation

Remove to fresh air. If not breathing, give artificial respiration. Call a physician or poison control center immediately. Do not use mouth-to-mouth method if victim ingested or inhaled the substance; give artificial respiration with the aid of a pocket mask equipped with a one-way valve or other proper respiratory medical device.

#### Ingestion

Immediate medical attention is required. Do NOT induce vomiting. Drink plenty of water.

Never give anything by mouth to an unconscious person.

|                                        |                                                                                                                                                                                                                                                                                                 |
|----------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Most important symptoms/effects</b> | Causes burns by all exposure routes. Product is a corrosive material. Use of gastric lavage or emesis is contraindicated. Possible perforation of stomach or esophagus should be investigated: Ingestion causes severe swelling, severe damage to the delicate tissue and danger of perforation |
| <b>Notes to Physician</b>              | Treat symptomatically                                                                                                                                                                                                                                                                           |

## 5. Fire-fighting measures

|                                         |                                                                                                                                                                           |
|-----------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Suitable Extinguishing Media</b>     | Use extinguishing measures that are appropriate to local circumstances and the surrounding environment. CO <sub>2</sub> , dry chemical, dry sand, alcohol-resistant foam. |
| <b>Unsuitable Extinguishing Media</b>   | No information available                                                                                                                                                  |
| <b>Flash Point</b>                      | No information available                                                                                                                                                  |
| <b>Method -</b>                         | No information available                                                                                                                                                  |
| <b>Autoignition Temperature</b>         | No information available                                                                                                                                                  |
| <b>Explosion Limits</b>                 |                                                                                                                                                                           |
| <b>Upper</b>                            | No data available                                                                                                                                                         |
| <b>Lower</b>                            | No data available                                                                                                                                                         |
| <b>Oxidizing Properties</b>             | Oxidizer                                                                                                                                                                  |
| <b>Sensitivity to Mechanical Impact</b> | No information available                                                                                                                                                  |
| <b>Sensitivity to Static Discharge</b>  | No information available                                                                                                                                                  |

### Specific Hazards Arising from the Chemical

The product causes burns of eyes, skin and mucous membranes. Oxidizer: Contact with combustible/organic material may cause fire. May ignite combustibles (wood paper, oil, clothing, etc.).

### Hazardous Combustion Products

Fumes of aluminum or aluminum oxide. Hydrogen chloride.

### Protective Equipment and Precautions for Firefighters

As in any fire, wear self-contained breathing apparatus pressure-demand, MSHA/NIOSH (approved or equivalent) and full protective gear. Thermal decomposition can lead to release of irritating gases and vapors.

### NFPA

**Health**  
3

**Flammability**  
0

**Instability**  
0

**Physical hazards**  
OX

## 6. Accidental release measures

|                                             |                                                                                                                                                                                                                                      |
|---------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Personal Precautions</b>                 | Use personal protective equipment as required. Evacuate personnel to safe areas. Avoid contact with skin, eyes or clothing.                                                                                                          |
| <b>Environmental Precautions</b>            | Should not be released into the environment. Do not allow material to contaminate ground water system.                                                                                                                               |
| <b>Methods for Containment and Clean Up</b> | Sweep up and shovel into suitable containers for disposal. Avoid dust formation. Soak up with inert absorbent material. Keep in suitable, closed containers for disposal. Sweep up and shovel into suitable containers for disposal. |

## 7. Handling and storage

|                 |                                                                                                                                                                                                                                                                                         |
|-----------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Handling</b> | Wear personal protective equipment/face protection. Do not get in eyes, on skin, or on clothing. Use only under a chemical fume hood. Do not breathe dust. Do not ingest. If swallowed then seek immediate medical assistance. Keep away from clothing and other combustible materials. |
| <b>Storage.</b> | Do not store near combustible materials. Keep containers tightly closed in a dry, cool and                                                                                                                                                                                              |

well-ventilated place. Corrosives area. Incompatible Materials. Strong reducing agents. Combustible material. Oxidizing agent.

## 8. Exposure controls / personal protection

### Exposure Guidelines

This product does not contain any hazardous materials with occupational exposure limits established by the region specific regulatory bodies.

### Engineering Measures

Ventilation systems. Ensure that eyewash stations and safety showers are close to the workstation location. Ensure adequate ventilation, especially in confined areas. Wherever possible, engineering control measures such as the isolation or enclosure of the process, the introduction of process or equipment changes to minimise release or contact, and the use of properly designed ventilation systems, should be adopted to control hazardous materials at source

### Personal protective equipment

#### **Eye Protection** **Hand Protection**

Goggles  
Protective gloves

| Glove material | Breakthrough time | Glove thickness | Glove comments         |
|----------------|-------------------|-----------------|------------------------|
| Natural rubber | See manufacturers | -               | Splash protection only |
| Nitrile rubber | recommendations   |                 |                        |
| Neoprene       |                   |                 |                        |
| PVC            |                   |                 |                        |

Inspect gloves before use. observe the instructions regarding permeability and breakthrough time which are provided by the supplier of the gloves. (Refer to manufacturer/supplier for information) gloves are suitable for the task: Chemical compatability, Dexterity, Operational conditions, User susceptibility, e.g. sensitisation effects, also take into consideration the specific local conditions under which the product is used, such as the danger of cuts, abrasion. gloves with care avoiding skin contamination.

#### **Respiratory Protection**

When workers are facing concentrations above the exposure limit they must use appropriate certified respirators. Follow the OSHA respirator regulations found in 29 CFR 1910.134 or European Standard EN 149. Use a NIOSH/MSHA or European Standard EN 149 approved respirator if exposure limits are exceeded or if irritation or other symptoms are experienced.

To protect the wearer, respiratory protective equipment must be the correct fit and be used and maintained properly

**Recommended Filter type:** Particulates filter conforming to EN 143

When RPE is used a face piece Fit Test should be conducted

### Environmental exposure controls

No information available.

### Hygiene Measures

Handle in accordance with good industrial hygiene and safety practice. Keep away from food, drink and animal feeding stuffs. Do not eat, drink or smoke when using this product. Remove and wash contaminated clothing and gloves, including the inside, before re-use. Wash hands before breaks and after work.

## 9. Physical and chemical properties

|                            |                          |
|----------------------------|--------------------------|
| <b>Physical State</b>      | Solid                    |
| <b>Appearance</b>          | Colorless                |
| <b>Odor</b>                | Odorless                 |
| <b>Odor Threshold</b>      | No information available |
| <b>pH</b>                  | 0-3 5% aq. sol           |
| <b>Melting Point/Range</b> | No data available        |
| <b>Boiling Point/Range</b> | No information available |
| <b>Flash Point</b>         | No information available |
| <b>Evaporation Rate</b>    | Not applicable           |

|                                        |                          |
|----------------------------------------|--------------------------|
| Flammability (solid,gas)               | No information available |
| Flammability or explosive limits       |                          |
| Upper                                  | No data available        |
| Lower                                  | No data available        |
| Vapor Pressure                         | No information available |
| Vapor Density                          | Not applicable           |
| Specific Gravity                       | No information available |
| Solubility                             | No information available |
| Partition coefficient; n-octanol/water | No data available        |
| Autoignition Temperature               | No information available |
| Decomposition Temperature              | No information available |
| Viscosity                              | Not applicable           |
| Molecular Formula                      | Al Cl3 O12 . 9 H2 O      |
| Molecular Weight                       | 487.46                   |

## 10. Stability and reactivity

|                                  |                                                                                                     |
|----------------------------------|-----------------------------------------------------------------------------------------------------|
| Reactive Hazard                  | Yes                                                                                                 |
| Stability                        | Stable under normal conditions. Oxidizer: Contact with combustible/organic material may cause fire. |
| Conditions to Avoid              | Incompatible products. Excess heat. Combustible material.                                           |
| Incompatible Materials           | Strong reducing agents, Combustible material, Oxidizing agent                                       |
| Hazardous Decomposition Products | Fumes of aluminum or aluminum oxide, Hydrogen chloride                                              |
| Hazardous Polymerization         | Hazardous polymerization does not occur.                                                            |
| Hazardous Reactions              | None under normal processing.                                                                       |

## 11. Toxicological information

### Acute Toxicity

Product Information No acute toxicity information is available for this product

### Component Information

Toxicologically Synergistic Products No information available

### Delayed and immediate effects as well as chronic effects from short and long-term exposure

Irritation No information available

Sensitization No information available

Carcinogenicity The table below indicates whether each agency has listed any ingredient as a carcinogen.

| Component                        | CAS-No     | IARC       | NTP        | ACGIH      | OSHA       | Mexico     |
|----------------------------------|------------|------------|------------|------------|------------|------------|
| Aluminum perchlorate nonahydrate | 81029-06-3 | Not listed | Not listed | Not listed | Not listed | Not listed |
| Perchloric acid, aluminum salt   | 14452-39-2 | Not listed | Not listed | Not listed | Not listed | Not listed |

Mutagenic Effects No information available

Reproductive Effects No information available.

Developmental Effects No information available.

Teratogenicity No information available.

|                                            |                                                                                                                                                                                                                                                            |
|--------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| STOT - single exposure                     | Respiratory system                                                                                                                                                                                                                                         |
| STOT - repeated exposure                   | None known                                                                                                                                                                                                                                                 |
| Aspiration hazard                          | No information available                                                                                                                                                                                                                                   |
| Symptoms / effects, both acute and delayed | Product is a corrosive material. Use of gastric lavage or emesis is contraindicated. Possible perforation of stomach or esophagus should be investigated: Ingestion causes severe swelling, severe damage to the delicate tissue and danger of perforation |
| Endocrine Disruptor Information            | No information available                                                                                                                                                                                                                                   |
| Other Adverse Effects                      | The toxicological properties have not been fully investigated.                                                                                                                                                                                             |

## 12. Ecological information

### Ecotoxicity

Do not empty into drains.

|                               |                           |
|-------------------------------|---------------------------|
| Persistence and Degradability | No information available  |
| Bioaccumulation/ Accumulation | No information available. |
| Mobility                      | No information available. |

## 13. Disposal considerations

|                        |                                                                                                                                                                                                                                                                 |
|------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Waste Disposal Methods | Chemical waste generators must determine whether a discarded chemical is classified as a hazardous waste. Chemical waste generators must also consult local, regional, and national hazardous waste regulations to ensure complete and accurate classification. |
|------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|

## 14. Transport information

### DOT

|               |        |
|---------------|--------|
| UN-No         | UN1481 |
| Hazard Class  | 5.1    |
| Packing Group | II     |

### TDG

|               |        |
|---------------|--------|
| UN-No         | UN1481 |
| Hazard Class  | 5.1    |
| Packing Group | II     |

### IATA

|                      |                                 |
|----------------------|---------------------------------|
| UN-No                | UN1481                          |
| Proper Shipping Name | PERCHLORATES, INORGANIC, N.O.S. |
| Hazard Class         | 5.1                             |
| Packing Group        | II                              |

### IMDG/IMO

|                      |                                 |
|----------------------|---------------------------------|
| UN-No                | UN1481                          |
| Proper Shipping Name | PERCHLORATES, INORGANIC, N.O.S. |
| Hazard Class         | 5.1                             |
| Packing Group        | II                              |

## 15. Regulatory information

All of the components in the product are on the following Inventory lists: X = listed U.S.A. (TSCA) Canada (DSL/NDL) Europe (EINECS/ELINCS/NLP) Australia (AICS) Korea (KECL) China (IECSC) Japan (ENCS) Philippines (PICCS) Taiwan (TCSI) Japan (ISHL) New Zealand (NZIoC) Japan (ISHL)

### International Inventories

| Component | CAS-No | DSL | NDL | TSCA | TSCA Inventory notification - Active-Inactive | EINECS | ELINCS | NLP |
|-----------|--------|-----|-----|------|-----------------------------------------------|--------|--------|-----|
|           |        |     |     |      |                                               |        |        |     |

|                                  |            |   |   |   |        |           |   |   |
|----------------------------------|------------|---|---|---|--------|-----------|---|---|
| Aluminum perchlorate nonahydrate | 81029-06-3 | - | - | - | -      | -         | - | - |
| Perchloric acid, aluminum salt   | 14452-39-2 | - | X | X | ACTIVE | 238-437-6 | - | - |

| Component                        | CAS-No     | IECSC | KECL     | ENCS | ISHL | TCSI | AICS | NZIoC | PICCS |
|----------------------------------|------------|-------|----------|------|------|------|------|-------|-------|
| Aluminum perchlorate nonahydrate | 81029-06-3 | -     | -        | -    | X    | X    | -    | -     | -     |
| Perchloric acid, aluminum salt   | 14452-39-2 | -     | KE-28138 | -    | -    | X    | -    | -     | -     |

**Legend:**

X - Listed '-' - Not Listed

**KECL** - NIER number or KE number (<http://ncis.nier.go.kr/en/main.do>)**DSL/NDL** - Canadian Domestic Substances List/Non-Domestic Substances List**TSCA** - United States Toxic Substances Control Act Section 8(b) Inventory**EINECS/ELINCS** - European Inventory of Existing Commercial Chemical Substances/EU List of Notified Chemical Substances**IECSC** - Chinese Inventory of Existing Chemical Substances**KECL** - Korean Existing and Evaluated Chemical Substances**ENCS** - Japanese Existing and New Chemical Substances**AICS** - Australian Inventory of Chemical Substances**PICCS** - Philippines Inventory of Chemicals and Chemical Substances**Canada**

SDS in compliance with provisions of information as set out in Canadian Standard - Part 4, Schedule 1 and 2 of the Hazardous Products Regulations (HPR) and meets the requirements of the HPR (Paragraph 13(1)(a) of the Hazardous Products Act (HPA)).

**Other International Regulations****Authorisation/Restrictions according to EU REACH**

Not applicable

**Safety, health and environmental regulations/legislation specific for the substance or mixture**

| Component                        | CAS-No     | OECD HPV       | Persistent Organic Pollutant | Ozone Depletion Potential | Restriction of Hazardous Substances (RoHS) |
|----------------------------------|------------|----------------|------------------------------|---------------------------|--------------------------------------------|
| Aluminum perchlorate nonahydrate | 81029-06-3 | Not applicable | Not applicable               | Not applicable            | Not applicable                             |
| Perchloric acid, aluminum salt   | 14452-39-2 | Not applicable | Not applicable               | Not applicable            | Not applicable                             |

| Component                        | CAS-No     | Seveso III Directive (2012/18/EC) - Qualifying Quantities for Major Accident Notification | Seveso III Directive (2012/18/EC) - Qualifying Quantities for Safety Report Requirements | Rotterdam Convention (PIC) | Basel Convention (Hazardous Waste) |
|----------------------------------|------------|-------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------|----------------------------|------------------------------------|
| Aluminum perchlorate nonahydrate | 81029-06-3 | Not applicable                                                                            | Not applicable                                                                           | Not applicable             | Not applicable                     |
| Perchloric acid, aluminum salt   | 14452-39-2 | Not applicable                                                                            | Not applicable                                                                           | Not applicable             | Not applicable                     |

**16. Other information****Prepared By**

Product Safety Department  
 Email: [chem.techinfo@thermofisher.com](mailto:chem.techinfo@thermofisher.com)  
[www.thermofisher.com](http://www.thermofisher.com)

**Revision Date**

29-March-2024

**Print Date**

29-March-2024

**Revision Summary**

New emergency telephone response service provider.

**Disclaimer**

The information provided in this Safety Data Sheet is correct to the best of our knowledge, information and belief at the

date of its publication. The information given is designed only as a guidance for safe handling, use, processing, storage, transportation, disposal and release and is not to be considered a warranty or quality specification. The information relates only to the specific material designated and may not be valid for such material used in combination with any other materials or in any process, unless specified in the text

**End of SDS**